

Level: Bachelor
Programme: BE
Course: Microprocessor

Semester: Fall

Year : 2019
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

7. Write short notes on: (Any two)
- Explain the instruction : MVI, ANI, CMP, JNC, NOP
 - RISC and CISC
 - RS232 Standard

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- Explain the Instruction execution Cycle of microprocessor. 5
 - Compare Von-Neumann's Architecture with Harvard architecture. 5
 - What are Flags? Write the function of all the flags present in 8085 microprocessor. 5
- What are the different flags of 8085 microprocessor? Explain the addressing modes of 8085 microprocessor with examples of each. 2+6
 - Write an ALP in 8085 to perform the following addition: 7

$$1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 + 9^2$$
- Define T-state. Draw the neat and labeled timing diagram for instruction MOV M, B. 8
 - Write the function of DMA with suitable diagram and explain the execution of DMA in Master mode and Slave mode. 7
- Differentiate between partial and complete address decoding. Design an address decoding circuits for two 16KB ROMs at address 2000H. 9
 - Explain Polled and Vectored interrupt with suitable block diagram. 6
- Draw the pin diagram of 8086 microprocessor and explain the different types of FLAGS of 8086. 7
 - Write an 8086 ALP for MASM in DOS mode to print each word of a string in different lines. 8
- What are the different pre-defined interrupts in 8086 microprocessor? Explain with the help of block diagram of 8086 IVT. 8
 - How can you transfer the data between two computers without using modem? Draw and explain the interface with handshake Signal. 7